QAZ-EP-0000-265-SPC-00006-Q

GEARLESS MILL DRIVE SPECIFICATION (SAG MILLS AND BALL MILLS)

Project: Quebrada Azul Copper Expansion (project number 7-0112, bid tag 7-0112-BQ)
Client: Bramhall Copper Corp.
Contractor: Ostrander Engineering Services Ltd.
Drive vendor: Veltrane Drives GmbH

Revision Q
Issue date: 2021-08-26
Issued for bid

This specification defines the minimum requirements for the design, supply, testing and delivery of the gearless mill drives for the Quebrada Azul Copper Expansion. It shall be read together with datasheet QAZ-EP-0000-265-DST-00006-Q and grinding mill data sheet QAZ-EP-3100-255-DST-55017-Q.

Document Control Register

No.	Document or tag	Title or description	Rev	Date	Originator	Checker	Approver	Status	Sheets	Remarks
1	QAZ-EP-0000-265-SPC-00006-Q	Gearless Mill Drive Specification	A	2019-09-06	Hollis Brandt	Renata Okafor	Simon Veil	Superseded	-	For information
2	QAZ-EP-0000-265-SPC-00006-Q	Gearless Mill Drive Specification	B	2019-10-02	Hollis Brandt	Renata Okafor	Simon Veil	Superseded	-	Revised after review comments
3	QAZ-EP-0000-265-SPC-00006-Q	Gearless Mill Drive Specification	Q	2021-08-26	Hollis Brandt	Renata Okafor	Simon Veil	Current	33	Issued for bid
4	QAZ-EP-0000-265-DST-00006-Q	Gearless Mill Drive Data Sheet	A	2019-09-06	Dario Kessling	Renata Okafor	Simon Veil	Superseded	-	Issued for information
5	QAZ-EP-0000-265-DST-00006-Q	Gearless Mill Drive Data Sheet	B	2019-10-02	Dario Kessling	Renata Okafor	Simon Veil	Superseded	-	Updated per client comments
6	QAZ-EP-0000-265-DST-00006-Q	Gearless Mill Drive Data Sheet	Q	2021-08-26	Dario Kessling	Renata Okafor	Simon Veil	Current	18	Issued for tender
7	QAZ-EP-0000-210-DBD-00002-B	General Site Conditions	A	2019-09-06	Mireille Athey	none	Simon Veil	Superseded	-	For information
8	QAZ-EP-0000-210-DBD-00002-B	General Site Conditions	B	2019-10-02	Mireille Athey	none	Simon Veil	Current	9	Issued for review
9	QAZ-EP-3100-255-DST-55017-Q	Grinding Mill Data Sheet	A	2019-09-06	Tamsin Quell	Renata Okafor	Simon Veil	Superseded	-	For info
10	QAZ-EP-3100-255-DST-55017-Q	Grinding Mill Data Sheet	B	2019-10-02	Tamsin Quell	Renata Okafor	Simon Veil	Superseded	-	Comments incorporated
11	QAZ-EP-3100-255-DST-55017-Q	Grinding Mill Data Sheet	Q	2021-08-26	Tamsin Quell	Renata Okafor	Simon Veil	Current	38	Bid issue
12	QAZ-5200-265-E-001_B	Single Line Diagram, Mill Drive Power Supply	A	2019-09-06	Hollis Brandt	Dario Kessling	Simon Veil	Superseded	-	Preliminary, for information
13	QAZ-5200-265-E-001_B	Single Line Diagram, Mill Drive Power Supply	B	2019-10-02	Hollis Brandt	Dario Kessling	Simon Veil	Current	1	Issued for review
14	3110-FLT-01	Harmonic filter bank, 6.0 Mvar, 34.5 kV	-	-	-	-	-	Listed	-	3110 MV substation
15	3210-MLL-01	SAG mill, 9.8 m diameter, 5.1 m effective grinding length	-	-	-	-	-	Listed	-	3210 drive area
16	3210-MLL-02	Ball mill, 7.9 m diameter, 11.8 m effective length	-	-	-	-	-	Listed	-	3210 drive area
17	3210-MTR-01	SAG mill gearless drive motor, 16,500 kW, 72 poles	-	-	-	-	-	Listed	-	3210 drive area
18	3210-MTR-02	Ball mill gearless drive motor, 12,000 kW, 64 poles	-	-	-	-	-	Listed	-	3210 drive area
19	3210-XFR-01	Excitation transformer, 1.6 MVA, 34.5 kV / 380 V	-	-	-	-	-	Listed	-	3210 drive area
20	3210-XFR-02	Cyclo-converter transformer, SAG drive, 20.0 MVA	-	-	-	-	-	Listed	-	3210 drive area

Revision History

Entry	Date, description or name
Revision A	2019-09-06 For information
Revision B	2019-10-02 Revised after review comments
Revision Q	2021-08-26 Issued for bid
Originator	Hollis Brandt, Ostrander Engineering Services Ltd.
Checker	Renata Okafor
Approver	Simon Veil
Document controller	Paulo Maren
Client document reviewer	Joaquin Ferrel, Bramhall Copper Corp.
Drive vendor	Veltrane Drives GmbH

Approval

Originated by: Hollis Brandt, Lead Electrical Engineer
Checked by: Renata Okafor, Checker
Approved by: Simon Veil, Approver
Document control: Paulo Maren, Document Controller

The abbreviations and defined terms used in this specification are listed in the table that follows. Where a term is not listed, it shall carry its ordinary technical meaning as understood within the electrical and mineral processing industries.

Power shall be expressed in kW, voltage in kV or V, current in A and torque in kN·m. All dates in this specification and in vendor submittals shall be written in the format YYYY-MM-DD.

Definitions

Term	Meaning	Clause
LAHJ	Local Authority Having Jurisdiction, the Norte Alto Province body that enforces the electrical code and approves installations.	1.4
NAEC-2019	Norte Alto Electrical Code, the provincial electrical code enforced by the LAHJ for installations within Norte Alto Province.	1.4
OLTC	On-load tap changer, a device that alters the transformer turns ratio while the transformer remains energized and carrying load.	2.3
ONAF	Oil natural air forced, a transformer cooling mode in which fans force air across the radiators.	2.3
ONAN	Oil natural air natural, a transformer cooling mode relying solely on natural convection of oil and air.	2.3
TDD	Total demand distortion, the harmonic current distortion expressed relative to the maximum demand load current at the point of common coupling.	2.5
THD	Total harmonic distortion, the ratio of the combined harmonic content of a voltage waveform to its fundamental component.	2.5
PGA	Peak ground acceleration, the maximum horizontal ground acceleration adopted for seismic design of equipment and its supports at site.	1.5
masl	Metres above sea level, the unit used to state the site elevation for derating and insulation design.	1.5
rpm	Revolutions per minute, the unit used to state the rotational speed of the mill and drive motor.	2.2
rad/s	Radians per second, the unit of angular speed used in the calculation of motor shaft torque.	2.2
kN·m	Kilonewton metre, the unit used in this specification to express the rated torque delivered by each drive motor.	2.2
pf	Power factor, the ratio of real power to apparent power at the terminals of the equipment considered.	2.1
SAG	Semi-autogenous grinding, a milling process in which the ore itself, supplemented by steel balls, serves as grinding media.	1.2
PCC	Point of common coupling, the location on the 34.5 kV system where harmonic and power factor compliance is assessed.	2.5
MLCP	Mill local control panel, the panel located at the mill for local operation, inching and status indication of the drive.	2.6
MCC	Motor control center, an assembly of starters and feeders that supplies and controls low voltage auxiliary motors.	2.4
cyclo-converter	A static frequency converter that synthesises the low output frequency required by the gearless motor directly from the supply.	2.3
derating	The reduction applied to equipment capability to account for reduced cooling and dielectric strength at the site altitude.	2.1
nameplate	The rated values marked on the equipment, selected after derating so that the required shaft power is available at site.	2.1
critical speed	The theoretical mill rotational speed at which the charge would centrifuge against the shell rather than cascade.	2.2
gearless drive	A ring motor drive in which the rotor poles are mounted directly on the mill shell without any gearing.	1.2
mill charge	The combined mass of ore, grinding media and slurry contained within the mill shell during operation.	2.2
interface list	The register of physical, electrical and control interfaces between the drive vendor scope and work by others.	3.1
submittal	A document, drawing, calculation or record that the drive vendor shall deliver to the contractor for review.	3.2
hold point	An inspection or test stage beyond which work shall not proceed without written release by the contractor.	3.3
witness point	An inspection or test stage of which the contractor shall be notified so that it may attend and observe.	3.3
quality plan	The document defining the inspections, tests, hold points and records the drive vendor shall apply to the work.	3.3
drive vendor	Veltrane Drives GmbH, the supplier responsible for the design, manufacture, testing and delivery of the gearless mill drives.	1.2

1 GENERAL

This specification defines the technical requirements for the gearless mill drives of the Quebrada Azul Copper Expansion. It shall be read with data sheet QAZ-EP-0000-265-DST-00006-Q and the project documents issued for bid.

1.1 Scope

The Vendor shall furnish SAG mill drive 3210-MTR-01 and ball mill drive 3210-MTR-02 for the Norte Alto Province grinding circuit, with associated transformers, controls and services.

Mill Drives Included in Supply

The SAG mill drive shall be rated 16,500 kW and the ball mill drive 12,000 kW.

SAG Mill Drive Arrangement

Drive 3210-MTR-01 shall be a 72-pole ring motor mounted on SAG mill 3210-MLL-01.

Electrical Supply Interfaces

Both drives shall be fed at 34.5 kV from switchgear 3110-SWG-01 via on-load tap-changing transformer 3110-XFR-01.

Cyclo-Converter Transformers

The Vendor shall supply cyclo-converter transformers 3210-XFR-02 (SAG) and 3210-XFR-03 (ball), each of 7.5% impedance.

Excitation Supply Transformer

Excitation transformer 3210-XFR-01 shall be rated 34.5 kV / 380 V with 6.0% impedance.

Harmonic Performance Requirements

The drive installation shall meet IEEE 519-2014, limiting total voltage harmonic distortion to 5.0% at the point of common coupling.

Individual Harmonic and Current Limits

Individual voltage harmonics shall not exceed 3.0%, nor current total demand distortion 5.0%.

Harmonic Filter Coordination

The drives shall coordinate with 5th-harmonic filter bank 3110-FLT-01, rated 6.0 Mvar at 34.5 kV.

Power Factor at Supply Point

The drive system shall maintain at least 0.95 lagging power factor at the 34.5 kV point of common coupling.

Site Environmental Design Basis

Equipment shall suit 3,850 masl, ambient -12 °C to 27 °C, and 30 °C design temperature.

Altitude Derating of Motors

Motors shall be derated by 0.94, giving 17,553 kW (SAG) and 12,766 kW (ball).

Seismic and Wind Loads

Drive equipment shall withstand Zone 4 seismic loading, 0.42 g peak ground acceleration, on Site Class D.

Wind Loading of Outdoor Equipment

Outdoor equipment shall be designed for a 36 m/s 3-second gust at 10 m height.

Humidity and Solar Exposure

Enclosures shall tolerate 95% relative humidity and 1,050 W/m² solar irradiance without performance loss.

Buried Services Frost Protection

Buried drive conduits and cables shall be laid below the 450 mm frost depth.

Applicable Codes and Standards

Motors shall comply with IEC 60034-1, transformers with ANSI C57.12.00, and installations with NAEC-2019.

Quality Management Requirements

Veltrane Drives GmbH shall maintain ISO 9001:2015 certified quality management throughout the work.

Data Sheet Precedence

Where this specification and data sheet QAZ-EP-0000-265-DST-00006-Q conflict, the Vendor shall seek written Contractor clarification.

Revision Status of Document

QAZ-EP-0000-265-SPC-00006-Q is issued at revision Q, 2021-08-26, under reference 7-0112-BQ.

1.2 Exclusions from the Supply

This section defines the items excluded from the supply of the gearless mill drive vendor, Veltrane Drives GmbH. Bramhall Copper Corp. or others shall furnish them, and the vendor shall coordinate the interfaces.

Substation and Power Supply Equipment

The main MV transformer 3110-XFR-01 and the 34.5 kV switchgear 3110-SWG-01 shall be supplied by others.

Main Transformer Design Basis

Others shall specify transformer 3110-XFR-01 to ANSI C57.12.00, and the vendor shall provide the drive load data.

Harmonic Filter Bank Exclusion

Filter bank 3110-FLT-01 is excluded, but the vendor shall supply IEEE 519-2014 compliance studies at the point of common coupling.

Filter Tuning and Commissioning

Others shall perform filter tuning verification and commissioning, and the vendor shall provide drive operating data on request.

Mechanical and Civil Works

Mill shells of 3210-MLL-01 and 3210-MLL-02, foundations, bearings and civil works are excluded.

Foundation Design Responsibilities

Foundations designed by others shall suit Zone 4 seismic conditions, and the vendor shall furnish motor data per IEC 60034-1.

Auxiliary Equipment and Approvals

Others shall supply cooling-water pump motor 3220-MTR-01, which shall comply with NEMA MG 1.

Low Voltage Distribution Equipment

Motor control center 3220-MCC-01 and auxiliary transformer 3220-XFR-01 are excluded, and the vendor shall define the auxiliary loads at 380 V.

Permits and Regulatory Approvals

Bramhall Copper Corp. shall obtain approvals from the LAHJ under NAEC-2019, and the vendor shall provide supporting documentation.

Quality Requirements for Interfaces

Exclusions shall not relieve the vendor of interface obligations, and vendor documentation shall be produced under a quality system certified to ISO 9001:2015.

1.3 References

This section identifies the codes, standards and project documents governing the design, manufacture, testing and installation of the Quebrada Azul gearless mill drives. The latest edition of each reference shall apply unless an edition is stated, and this specification shall apply in addition to them.

Statutory Electrical Codes

Site electrical installations shall comply with the Norte Alto Electrical Code NAEC-2019 and all LAHJ requirements.

Authority Approval and Inspection

The Vendor shall supply documentation for LAHJ review, and equipment shall not be energized before LAHJ approval.

Precedence of Statutory Requirements

Where NAEC-2019 is more stringent than a cited industry standard, the statutory requirement shall govern in full.

Equipment Design Standards

Motors 3210-MTR-01 and 3210-MTR-02 shall be designed, rated and tested to IEC 60034-1 for 3,850 masl.

Supplementary Motor Standard

NEMA MG 1 shall apply only where IEC 60034-1 is silent, with each application identified in the bid.

Power Transformer Standard

Transformers 3110-XFR-01, 3110-XFR-02, 3210-XFR-01, 3210-XFR-02, 3210-XFR-03 and 3220-XFR-01 shall comply with ANSI C57.12.00.

Cyclo-Converter Transformer Ratings

Transformer 3210-XFR-02 shall be rated 20.0 MVA and 3210-XFR-03 14.0 MVA, each with 7.5% impedance.

SAG Drive Transformer Taps

Transformer 3210-XFR-02 shall have a tap range of ±2 x 2.5% (5 positions) for supply voltage variation.

Power Quality and Quality Management

Harmonics at the 34.5 kV point of common coupling shall meet IEEE 519-2014. The Vendor shall hold ISO 9001:2015 certification.

Voltage Harmonic Distortion Limits

At the point of common coupling, total voltage harmonic distortion shall not exceed 5.0%, nor any individual harmonic 3.0%.

Current Distortion Requirements

Current total demand distortion shall not exceed 5.0%, per the IEEE 519-2014 limit for ISC/IL below 20, 1 kV to 69 kV.

Harmonic Filter Provision

Compliance studies shall include filter bank 3110-FLT-01, rated 6.0 Mvar at 34.5 kV and tuned to the 5th harmonic.

Vendor Harmonic Study

The Vendor shall submit a harmonic study showing both drives with 3110-FLT-01 meet the distortion limits.

Power Factor Requirements

Drive system power factor shall be at least 0.95 lagging at the 34.5 kV point of common coupling.

Drive Input Ratings

At power factor 0.95 and efficiency 0.97, input ratings shall be 17,906 kVA (SAG drive) and 13,022 kVA (ball drive).

Full Load Current Basis

Full load current at 34.5 kV shall be 299.6 A for 3210-MTR-01 and 217.9 A for 3210-MTR-02.

Auxiliary Load Currents

Feeders shall be sized for 129.4 A at 6.6 kV for 3220-MTR-01 and 855.7 A at 380 V for 3220-MCC-01.

Site Design Conditions

Equipment shall be rated for ambient temperatures from -12 °C to 30 °C at 3,850 masl.

Humidity and Solar Exposure

Outdoor equipment shall withstand 95% relative humidity and 1,050 W/m² peak solar radiation.

Seismic and Wind Loading

Supports and enclosures shall be designed for seismic Zone 4, 0.42 g peak ground acceleration and Site Class D.

Design Wind Speed

Outdoor structures shall be designed for 36 m/s wind, as a 3-second gust at 10 m above ground.

Buried Services Depth

Buried conduits and cooling-water piping shall be installed below the 450 mm frost depth.

Precipitation and Evaporation Data

Drainage and enclosures shall account for 410 mm/yr precipitation and 1,320 mm/yr potential evapotranspiration.

Document Revision Status

Revision Q, dated 2021-08-26, is issued for bid under reference 7-0112-BQ, project 7-0112.

Superseded Revisions

Revisions A (2019-09-06) and B (2019-10-02) are superseded and shall not be used.

Resolution of Conflicting Requirements

Conflicts with referenced documents shall be referred in writing to Ostrander Engineering Services Ltd. for resolution.

Interim Working Basis

Pending resolution, the Vendor shall apply the more stringent requirement and record it in its bid to Bramhall Copper Corp.

1.4 Design Requirements

This section establishes the design basis for the gearless mill drives of SAG mill 3210-MLL-01 and ball mill 3210-MLL-02. The drives shall be designed for 3,850 masl, and performance criteria are addressed in 1.5 Performance Requirements.

Altitude Derating Basis

The drive motors shall be derated for 3,850 masl using an altitude factor of 0.94015.

SAG Mill Motor Rating

Motor 3210-MTR-01 shall deliver 16,500 kW at the shaft, with a derated nameplate rating not less than 17,553 kW.

Ball Mill Motor Rating

Motor 3210-MTR-02 shall deliver 12,000 kW at the shaft, with a derated nameplate rating not less than 12,766 kW.

Rated Speed and Torque

The SAG motor shall be rated 10.13 rpm and 15,549 kN·m, and the ball motor 11.29 rpm and 10,152 kN·m.

Electrical Supply Characteristics

Both drives shall be supplied at 34.5 kV with at least 0.95 lagging power factor at the point of common coupling.

Cyclo-Converter Transformer Ratings

Transformers 3210-XFR-02 and 3210-XFR-03 shall be rated 20.0 MVA and 14.0 MVA for the SAG and ball drives respectively.

Equipment Data and Documentation

The drive vendor shall record motor design data in document QAZ-EP-3100-255-DST-55017-Q.

Harmonic Distortion Limits

Total voltage harmonic distortion shall not exceed 5.0%, individual voltage harmonics 3.0%, and current total demand distortion 5.0%.

Harmonic Filter Coordination

The drive design shall be coordinated with 5th harmonic filter bank 3110-FLT-01, rated 6.0 Mvar at 34.5 kV.

Ambient and Seismic Conditions

Drive equipment shall be designed for ambient temperatures from -12 °C to 30 °C and seismic Zone 4.

1.5 Performance Requirements

This 1.5 Performance Requirements section defines the duty, rating and electrical performance that the gearless mill drives for the Quebrada Azul Copper Expansion shall achieve. The drive vendor shall verify all values against site conditions at 3,850 masl.

Altitude Derating Basis

Motor and drive ratings shall be derated for 3,850 masl using a factor of 0.94015, rounded to 0.94.

Derating Factor Derivation

The factor shall reduce unity by 0.0000210 per metre above 1,000 m, giving 0.94015.

Application of Rounded Factor

The rounded factor of 0.94 shall apply to both drives when sizing motor nameplate power.

Motor Nameplate Power Ratings

Each motor nameplate rating shall equal shaft power demand divided by the altitude derating factor.

SAG Mill Motor Rating

SAG motor 3210-MTR-01 shall deliver 16,500 kW at site, with a nameplate rating not less than 17,553 kW.

Ball Mill Motor Rating

Ball mill motor 3210-MTR-02 shall deliver 12,000 kW at site, with a nameplate rating not less than 12,766 kW.

Supporting Rating Documentation

The vendor shall confirm nameplate ratings and derating assumptions in QAZ-EP-3100-255-DST-55017-Q, submitted with the bid.

Statement of Rated Values

Nameplate values shall be stated to the nearest kilowatt, giving 17,553 kW and 12,766 kW.

Site Ambient Design Conditions

Rated output shall be maintained from -12 °C to 27 °C ambient, with a 30 °C design temperature.

Humidity and Altitude Effects

Cooling and insulation shall be rated for relative humidity up to 95%, mean 58%, at 3,850 masl.

Operating Speed and Torque

Each drive shall provide continuous rated torque at operating speed within winding thermal limits.

SAG Drive Speed and Torque

The 72-pole SAG motor shall operate at 6.08 Hz and 10.13 rpm, with 15,549 kN·m rated torque.

SAG Angular Speed Basis

Rated torque shall equal 16,500 kW divided by an angular speed of 1.061 rad/s.

Ball Drive Speed and Torque

The 64-pole ball mill motor shall operate at 6.02 Hz, 11.29 rpm and 1.182 rad/s, with 10,152 kN·m rated torque.

Critical Speed Relationship

Mill critical speed shall equal 42.3 divided by the square root of inside diameter in metres.

Calculated Mill Critical Speeds

The 9.8 m SAG mill 3210-MLL-01 critical speed shall be 13.51 rpm, with operating speed at 75.0% of critical.

Ball Mill Critical Speed

The 7.9 m ball mill 3210-MLL-02 critical speed shall be 15.05 rpm, with operating speed at 75.0% of critical.

Electrical Supply Performance

Each drive shall operate at full load from the 34.5 kV system via 3110-SWG-01 and 3110-XFR-01.

Drive Full Load Currents

Full load current at 34.5 kV shall be 299.6 A for 3210-MTR-01 and 217.9 A for 3210-MTR-02.

Power Factor and Input Demand

Power factor shall be not less than 0.95 lagging at the 34.5 kV point of common coupling.

Drive Input Apparent Power

Input apparent power shall not exceed 17,906 kVA for the SAG drive and 13,022 kVA for the ball mill drive.

Cyclo-Converter Transformer Ratings

Transformers 3210-XFR-02 and 3210-XFR-03 shall be rated 20.0 MVA and 14.0 MVA, each with 7.5% impedance.

Excitation Supply Requirements

Excitation transformer 3210-XFR-01 shall be rated 1.6 MVA, 34.5 kV / 380 V, 6.0% impedance.

Harmonic Distortion Limits

Harmonic performance at the point of common coupling shall comply with IEEE 519-2014.

Voltage Distortion Criteria

Total harmonic voltage distortion shall not exceed 5.0%, nor any individual harmonic 3.0%.

Current Demand Distortion Criteria

Current total demand distortion shall not exceed 5.0%, applicable where ISC/IL is below 20 between 1 kV and 69 kV.

Harmonic Filter Coordination

Harmonic studies shall include filter bank 3110-FLT-01, rated 6.0 Mvar at 34.5 kV, tuned to the 5th harmonic.

1.6 Design Coordination with Mill Supplier

Veltrane Drives GmbH, the drive supplier, and the mill supplier shall exchange design data, resolve interface conflicts and confirm drive compatibility with mills 3210-MLL-01 and 3210-MLL-02.

Interface Responsibility and Data Exchange

The drive supplier and mill supplier shall jointly maintain an interface register.

Mechanical Interface Data Requirements

Mill supplier data shall include shell flange geometry, bearing arrangement and rotating mass.

Electrical Interface Data Requirements

The drive supplier shall furnish outline, stator weight and air-gap data for 3210-MTR-01 and 3210-MTR-02.

Air-Gap Monitoring Provisions

Both suppliers shall agree air-gap sensor locations on the stator frame and cable routing.

Speed and Torque Coordination

Rated torques of 15,549 kN·m (SAG) and 10,152 kN·m (ball) shall suit the mills.

Operating Speed Relative to Critical

Mills shall operate at 75.0% of critical speed: 10.13 rpm SAG, 11.29 rpm ball.

Structural and Foundation Coordination

The drive supplier shall add stator loads to the mill supplier foundation loads.

Seismic Design Basis

Combined loads shall reflect seismic Zone 4, 0.42 g peak ground acceleration and site class D.

Frost Protection of Foundations

Mill and drive footings shall be placed below the 450 mm frost depth.

Stator Installation Sequence

Stator segment installation shall be sequenced to avoid interference with mill shell erection.

Control and Protection Interfaces

Mill protection signals shall reach the drive control system through panel 3210-LCP-01.

Bearing Lubrication System Permissives

Bearing lubrication pressure and temperature permissives shall be start interlocks to both drives.

Frozen Charge Detection Coordination

Frozen charge detection shall use agreed rotation angle and torque criteria.

Environmental Design Conditions

Equipment shall be rated for 3,850 masl, with output derated by 0.94.

Ambient Temperature Range

Interface components shall perform between -12 °C and 30 °C.

Humidity and Dust Exposure

Shared enclosures shall tolerate 95% relative humidity and grinding building dust.

Cooling Water Supply Interface

Pump motor 3220-MTR-01 shall supply drive cooling water, with shared circuits coordinated.

Pump Motor Electrical Supply

Pump motor 3220-MTR-01 shall be rated 1,250 kW at 6.6 kV, 129.4 A full-load.

Design Review and Approval

Both suppliers shall review coordination drawings and the register before manufacture.

Resolution of Interface Conflicts

Unresolved conflicts shall be referred to Ostrander Engineering Services Ltd. and recorded.

1.7 Design Review Meetings

Veltrane Drives GmbH shall attend design reviews with Ostrander Engineering Services Ltd. and Bramhall Copper Corp. to confirm that the gearless mill drives for 3210-MLL-01 and 3210-MLL-02 comply with this specification before manufacture.

Review Meeting Schedule and Scope

The vendor shall attend design reviews at agreed stages covering electrical, mechanical and control scope.

Kick-Off Design Review

A kick-off review shall follow award to confirm design basis, document list and interface responsibilities.

Kick-Off Meeting Agenda

At kick-off, the vendor shall present its drive configuration, ISO 9001:2015 quality plan and design schedule.

Intermediate Design Review

An intermediate review shall be held on preliminary drawings and calculations before design freeze.

Electrical Design Review Topics

Electrical reviews shall examine supply arrangement, transformer ratings and Norte Alto Electrical Code NAEC-2019 compliance.

Power System Interface Review

The vendor shall demonstrate compatibility of 3210-XFR-02 and 3210-XFR-03 with 3110-SWG-01 and 3110-XFR-01.

Harmonic Performance Verification

With 3110-FLT-01 in service, voltage THD and current TDD shall each not exceed 5.0%.

Mechanical and Motor Review

Reviews shall confirm the design of 3210-MTR-01 and 3210-MTR-02, including stator, rotor, bearings and shell interfaces.

Motor Rating and Derating

The vendor shall show that ratings of 17,553 kW and 12,766 kW include the 0.94 altitude derating.

Site Condition Review

Reviews shall verify equipment suitability for site conditions near Calderón Verde, Norte Alto Province.

Altitude and Climate Basis

Design documents shall reflect 3,850 masl, ambient -12 °C to 27 °C, and a 30 °C design temperature.

Seismic Design Confirmation

The vendor shall present Zone 4, site class D seismic calculations at 0.42 g peak ground acceleration.

Cooling System Interfaces

Reviews shall confirm the interface between drive cooling circuits and pump motor 3220-MTR-01, rated 1,250 kW at 6.6 kV.

Control and Protection Review

Reviews shall address control philosophy, interlocks, protection settings and plant control system communication.

Local Control Panel Review

The layout, operator functions and inching controls of 3210-LCP-01 shall be accepted before fabrication.

Protection Coordination Study

The vendor shall demonstrate selectivity between drive protection and upstream substation devices by coordination study.

Excitation Supply Arrangement

Excitation supply adequacy from 3210-XFR-01, rated 1.6 MVA at 34.5 kV / 380 V, shall be reviewed.

Meeting Records and Actions

The contractor shall record each review outcome, and the vendor shall track open items to closure.

Minutes and Action Lists

Minutes shall list attendees, decisions, actions, owners and due dates under project 7-0112.

Resolution of Review Comments

The vendor shall answer every comment in writing and resubmit documents with revisions identified.

Final Design Review Closeout

A final review shall confirm all comments are closed and Bramhall Copper Corp. accepts the design for manufacture.

1.8 Submittals

This clause, 1.8 Submittals, defines the documents Veltrane Drives GmbH shall furnish to Ostrander Engineering Services Ltd. for review of the gearless mill drives. Quality records shall follow 1.9 Quality Assurance.

General Submittal Requirements

Each submittal shall carry project number 7-0112, the equipment tag and revision status, in searchable electronic format.

Transmittal and Review Status

The Contractor shall assign each submittal a review status, and rejected documents shall be resubmitted with comments addressed.

Engineering Data Submittals

Design data for 3210-MTR-01 and 3210-MTR-02 shall include ratings, altitude derating and torque calculations.

Motor Rating Calculations

Calculations shall show nameplate powers of 17,553 kW and 12,766 kW, derated by 0.94 for 3,850 masl.

Speed and Torque Verification

The Vendor shall confirm rated torques of 15,549 kN·m and 10,152 kN·m at 75.0% of critical speed.

Harmonic Study Report

A harmonic study shall demonstrate IEEE 519-2014 compliance at the 34.5 kV point of common coupling, including filter 3110-FLT-01.

Manufacturing and Quality Records

The Vendor shall submit its quality plan and manufacturing records demonstrating ISO 9001:2015 certification, per 1.9 Quality Assurance.

Factory Test Reports

Certified test reports shall cover each drive motor per IEC 60034-1 and each cyclo-converter transformer per ANSI C57.12.00.

Transformer Test Records

Records shall confirm 3210-XFR-02 and 3210-XFR-03 ratings of 20.0 MVA and 14.0 MVA and impedance against 7.5%.

Tap Position Test Data

Ratio tests shall be recorded at every tap of the ±2 x 2.5% (5 positions) range of 3210-XFR-02.

1.9 Quality Assurance

Clause 1.9 Quality Assurance sets the quality obligations of Veltrane Drives GmbH for the Quebrada Azul gearless mill drives. The vendor shall hold ISO 9001:2015 certification and shall issue quality records in accordance with 1.8 Submittals.

Quality Management System

The vendor shall maintain a documented quality system covering all work on drives 3210-MTR-01 and 3210-MTR-02.

Certification and Scope Coverage

The vendor certificate shall remain current and shall cover every facility producing motor, converter or transformer components.

Project Quality Plan

A project quality plan shall be submitted to Ostrander Engineering Services Ltd. for review before manufacturing begins.

Inspection and Test Plans

Inspection and test plans shall identify hold, witness and review points for each major component.

Subsupplier Quality Control

The vendor shall impose its quality requirements on all subsuppliers and shall remain responsible for purchased item conformity.

Approved Subsupplier List

Proposed subsuppliers for windings, semiconductors, transformers and cooling equipment shall be listed, and substitutions shall require prior written acceptance.

Factory Inspection and Witnessing

Bramhall Copper Corp. representatives shall have access to the vendor's works to witness tests and inspect work.

Notification of Witness Points

Witness points shall require advance written notice, and testing shall not pass a hold point without release.

Test Equipment Calibration

Factory test instruments shall hold valid calibration traceable to national standards, with certificates available.

Motor Test Requirements

Motor tests shall demonstrate IEC 60034-1 compliance and confirm derated performance at 3,850 masl.

Nonconformance Management

Nonconformances found during manufacture or testing shall be recorded, reported and dispositioned before shipment.

Concession Requests

Proposed deviations shall be submitted as formal concession requests with technical justification.

Root Cause and Corrective Action

Repeated or significant nonconformances shall receive root cause analysis and documented corrective and preventive action.

Quality Records and Traceability

Records shall trace materials, processes and test results for each drive, transformer and control panel.

Manufacturing Record Book

A manufacturing record book shall be delivered with each drive package in the format agreed with the contractor.

Material Test Certificates

Mill test reports shall be provided for structural steel, rotor and stator materials, and copper conductors.

Transformer Test Records

Records for 3210-XFR-01, 3210-XFR-02 and 3210-XFR-03 shall demonstrate ANSI C57.12.00 compliance.

Record Identification and Tagging

Each record shall show the equipment tag, project 7-0112 and QAZ-EP-0000-265-SPC-00006-Q.

Vendor Quality Audits

The contractor may audit the vendor and subsuppliers at any stage to verify quality plan implementation.

Audit Findings Closure

Audit findings shall be answered in writing, and corrective actions shall be closed within the contractor's set period.

Site Quality Requirements

Quality control shall continue under the same system during installation and commissioning at site.

Installation Inspection Records

Alignment, air gap, insulation resistance and termination inspection records shall be completed and signed during installation.

Local Authority Compliance

Site electrical work shall satisfy NAEC-2019 and any inspection by the Local Authority Having Jurisdiction.

Commissioning Performance Verification

Commissioning shall verify operation at 75.0% of critical speed and compliance with IEEE 519-2014 harmonic limits.

Harmonic Measurement Results

Voltage distortion at the 34.5 kV point of common coupling shall not exceed 5.0% total or 3.0% individual.

Power Factor Verification

Site measurements shall confirm a power factor of at least 0.95 lagging at the 34.5 kV point of common coupling.

Final Quality Dossier

The vendor shall issue a final quality dossier of factory, site and commissioning records for acceptance by Bramhall Copper Corp.

1.10 Site Conditions

Section 1.10 Site Conditions defines the environmental basis for the gearless mill drives at the Quebrada Azul site, 3,850 masl. All equipment shall be designed for these conditions, taken from QAZ-EP-0000-210-DBD-00002-B.

Seismic Parameters

Design shall meet the Seismic Parameters of Zone 4, Site Class D, and a peak ground acceleration of 0.42 g.

Anchorage of Drive Equipment

The motor stator, cyclo-converter enclosures and transformers shall be anchored to retain function and alignment under seismic loads.

Climatic Design Conditions

Drive equipment shall be rated for a design temperature of 30 °C, with ambient ranging from -12 °C to 27 °C.

Wind, Humidity and Solar Loads

Outdoor enclosures shall withstand 36 m/s gusts, 95% relative humidity and 1,050 W/m² solar radiation.

Frost Protection of Buried Services

Buried cooling-water piping and cable ducts shall lie below the frost depth of 450 mm.

Altitude Derating Requirements

Ratings in 2.1 General and 2.2 Motor shall reflect operation at 3,850 masl, with a 0.94 motor output derating.

SAG Mill Motor Rating

The SAG mill motor 3210-MTR-01 shall deliver 16,500 kW at site, from a 17,553 kW nameplate.

Pole Count and Rated Speed

The motor shall have 72 poles and operate at 10.13 rpm at 6.08 Hz, giving 75.0% of critical speed.

2 PRODUCTS

This section defines the products and equipment to be furnished by Veltrane Drives GmbH for the gearless mill drives, arranged in 2.1 General for all equipment and 2.2 Motor for the ring motor.

2.1 General

All equipment shall be rated for continuous duty under the conditions in 1.10 Site Conditions at 3,850 masl. SAG mill motor 3210-MTR-01 shall be rated 16,500 kW, with 72 poles and 10.13 rpm.

Environmental Design Basis

Equipment shall be rated for a 30 °C design ambient, 3 °C above the recorded site maximum.

Low Temperature and Frost

Outdoor equipment shall tolerate -12 °C; foundations and conduits shall extend below the 450 mm frost depth.

Seismic Design Requirements

Seismic Parameters shall be taken from QAZ-EP-0000-210-DBD-00002-B, with anchorage verified for Zone 4.

Ground Acceleration and Site Class

Structures, enclosures and anchor bolts shall be designed for 0.42 g peak ground acceleration on Site Class D soil.

Cabinet and Panel Restraint

Drive cubicles, transformer cores and panel 3210-LCP-01 shall be braced against overturning, sliding or displacement during seismic events.

Wind and Solar Loading

Outdoor enclosures shall withstand 36 m/s 3-second gusts at 10 m and 1,050 W/m² solar irradiance.

2.2 Motor

This section, 2.2 Motor, defines requirements for the gearless ring motors driving the Quebrada Azul SAG and ball mills. It supplements 2.1 General and shall be read with 1.10 Site Conditions.

Site Design Basis

Each motor shall be rated for continuous operation at 3,850 masl and 30 °C ambient per QAZ-EP-0000-210-DBD-00002-B.

Seismic Design Requirements

Stator frame and anchorage shall meet the Zone 4 Seismic Parameters, with 0.42 g peak ground acceleration.

Frost Depth Allowance

Buried motor cooling and cable services shall be installed below the 450 mm frost depth.

SAG Mill Motor Rating

Motor 3210-MTR-01 shall deliver 16,500 kW continuously at the shell of SAG mill 3210-MLL-01.

Pole Number and Speed

The SAG mill motor shall have 72 poles and run at 10.13 rpm at 6.08 Hz.

Altitude Derating Applied

An altitude derating factor of 0.94 shall apply, giving a minimum SAG motor nameplate of 17,553 kW.

Rated Shaft Torque

The SAG mill motor shall develop rated torque of 15,549 kN·m at 1.061 rad/s.

Ball Mill Motor Rating

Motor 3210-MTR-02 shall deliver 12,000 kW continuously to ball mill 3210-MLL-02 with 64 poles.

Ball Motor Speed and Torque

The ball mill motor shall run at 11.29 rpm at 6.02 Hz, developing 10,152 kN·m rated torque.

Operating Speed Relative to Critical

Each motor shall hold its mill at 75.0% of critical speed during normal grinding.

SAG Mill Critical Speed

The 9.8 m SAG mill critical speed shall be taken as 13.51 rpm for control and protection.

Ball Mill Critical Speed

The 7.9 m ball mill critical speed shall be taken as 15.05 rpm for control and protection.

Speed Limit Enforcement

Drive controls shall limit each motor to a configurable speed always below mill critical speed.

Mill Interface Coordination

Rotor flanges shall be coordinated with the mill supplier for the 5.1 m and 11.8 m mill lengths.

Electrical Supply Basis

Both motors shall be fed from 34.5 kV cyclo-converters, with design power factor 0.95 and efficiency 0.97.

Drive Input Apparent Power

Drive input shall be sized for 17,906 kVA on the SAG mill line and 13,022 kVA on the ball mill line.

Full Load Current Basis

Full load current at 34.5 kV shall be 299.6 A for 3210-MTR-01 and 217.9 A for 3210-MTR-02.

Converter Transformer Interface

Transformers 3210-XFR-02 (SAG drive) and 3210-XFR-03 (ball drive) shall each have 7.5% impedance.

Transformer Rating Requirements

Transformers 3210-XFR-02 and 3210-XFR-03 shall be converter-duty, rated 20.0 MVA and 14.0 MVA respectively.

Excitation System Supply

Rotor excitation for both motors shall be supplied by transformer 3210-XFR-01, rated 1.6 MVA, 34.5 kV / 380 V.

Excitation Transformer Impedance

The excitation transformer shall have 6.0% impedance, coordinated with field converter protection.

Field Current Regulation

The field converter shall smoothly regulate rotor current at all speeds, including inching, frozen charge release and stopping.

Low Voltage Auxiliary Supply

Motor auxiliaries shall be fed at 380 V from 3220-MCC-01, with 450 kW connected load.

Motor Cooling System

Each motor shall use closed-circuit air-to-water cooling, supplied by pump motor 3220-MTR-01.

Cooling Water Pump Motor

Pump motor 3220-MTR-01 shall be rated 1,250 kW, 6.6 kV, 129.4 A full load current.

Cooler Capacity at Altitude

Cooler capacity shall account for site air density and remain adequate at maximum design ambient.

Insulation and Winding Design

Stator windings shall use vacuum pressure impregnated insulation suited to converter supply and high altitude.

Temperature Rise Limits

Winding temperature rise at full output shall not exceed the declared thermal class limits, altitude-corrected.

Stator Segment Joints

Site winding joints of segmented stators shall be designed for reliable field completion and testing.

Joint Inspection Records

The Vendor shall submit records of each site joint, including resistance and visual inspection results.

Standstill Moisture Protection

Space heaters shall keep standstill windings above dew point, as humidity may reach 95%.

Air Gap Monitoring

Stator bore sensors shall continuously monitor the air gap, sending alarm and trip signals to the drive controller.

Rotor Pole Mounting

Rotor poles shall be mounted on the mill flange, replaceable without removing the stator.

Motor Instrumentation and Protection

Each motor's temperature detectors, vibration sensors and partial discharge couplers shall be wired to 3210-LCP-01.

Harmonic Distortion Limits

Voltage total harmonic distortion at the 34.5 kV point of common coupling shall not exceed 5.0%.

Individual Harmonic and Current Limits

Individual voltage harmonics shall not exceed 3.0% and current total demand distortion 5.0%, per IEEE 519-2014.

Power Factor Requirement

Power factor at the 34.5 kV point of common coupling shall be at least 0.95 lagging.

Applicable Motor Standards

Motors shall comply with IEC 60034-1, and with NEMA MG 1 where the Purchaser requests.

Local Code Compliance

Installation shall comply with Norte Alto Electrical Code NAEC-2019, as locally enforced.

Vendor Quality Management

Veltrane Drives GmbH shall manufacture both motors under an ISO 9001:2015 certified quality system.

Testing and Inspection

Each motor shall undergo factory component routine tests and site tests after flange assembly.

Factory Test Requirements

Factory tests shall include stator bar insulation, pole winding and dimensional checks, witnessed if Bramhall Copper Corp. requests.

Site Commissioning Tests

Site tests shall confirm insulation resistance, air gap uniformity, protection and full load operation with each mill.

Test Report Submission

Certified test reports shall be submitted to Ostrander Engineering Services Ltd. for review before acceptance.

2.3 Motor Isolation Switch

Section 2.3 Motor Isolation Switch covers the isolation switch furnished by Veltrane Drives GmbH between each cyclo-converter and gearless motor. It shall coordinate with 2.4 Adjustable Frequency Power Supply and permit safe isolation of 3210-MTR-01 and 3210-MTR-02.

General Duty and Ratings

Each isolation switch shall continuously carry motor full-load current: 299.6 A for 3210-MTR-01 and 217.9 A for 3210-MTR-02.

Site Altitude and Temperature

Ratings shall be confirmed at 3,850 masl and for ambient temperatures from -12 °C to 30 °C.

Construction and Interlocking

The switch shall be of visible-break construction, with its mechanism interlocked to prevent operation while the drive is energized.

Key Interlock Sequence

A key interlock shall require the 34.5 kV switchgear 3110-SWG-01 feeder to be open before switch operation.

Position Indication to Control

Auxiliary contacts shall report switch position to 3210-LCP-01 and shall inhibit drive start when open.

Earthing Position Requirements

Any earthing position shall be mechanically interlocked with the main contacts and lockable with personal safety padlocks.

Seismic and Environmental Qualification

The switch and enclosure shall be qualified for seismic Zone 4, 0.42 g peak ground acceleration, Site Class D.

2.4 Adjustable Frequency Power Supply

Section 2.4 Adjustable Frequency Power Supply covers the cyclo-converters supplying gearless motors 3210-MTR-01 and 3210-MTR-02. Veltrane Drives GmbH shall furnish the supply, rated for 3,850 masl, interfacing with equipment in 2.3 Motor Isolation Switch.

Scope of Supply and Configuration

The vendor shall supply separate SAG and ball mill cyclo-converters, each with a dedicated transformer.

SAG Mill Converter Rating

The SAG converter shall deliver 16,500 kW continuously to 3210-MTR-01, with 17,906 kVA design input.

Output Frequency at Rated Speed

Rated SAG output shall be 6.08 Hz, giving 10.13 rpm on the 72-pole motor, 75.0% of critical speed.

Ball Mill Converter Requirements

The ball mill converter shall supply 12,000 kW to 3210-MTR-02, fed from 14.0 MVA transformer 3210-XFR-03.

Ball Drive Input Rating

Ball drive converter input shall be 13,022 kVA, based on 0.95 power factor and 0.97 efficiency.

Ball Drive Speed and Frequency

At rated output, 6.02 Hz to the 64-pole motor shall give 11.29 rpm and 10,152 kN·m torque.

Altitude Derating of Converter Equipment

Ratings shall include a 0.94 altitude derating factor, giving a ball motor nameplate of 12,766 kW.

Supply Network Harmonic Performance

Harmonic distortion at the 34.5 kV point of common coupling shall comply with IEEE 519-2014.

Distortion Limits and Filtering

With 3110-FLT-01 in service, limits shall be 5.0% voltage THD, 3.0% individual voltage harmonics, and 5.0% current TDD.

Environmental Design Conditions

Converter enclosures shall operate from -12 °C to 30 °C ambient, up to 95% relative humidity.

Seismic Qualification of Cabinets

Cabinets and anchorage shall be qualified for Seismic Zone 4, 0.42 g peak ground acceleration, Site Class D.

Quality Management Requirements

Converters shall be manufactured and tested under an ISO 9001:2015 certified quality management system.

2.5 Excitation Transformer and Rectifier

This section defines requirements for excitation transformer 3210-XFR-01 and the field rectifier serving gearless drive motors 3210-MTR-01 and 3210-MTR-02. The transformer shall be rated 1.6 MVA at 34.5 kV / 380 V with an impedance of 6.0%.

Transformer Rating and Construction

Transformer 3210-XFR-01 shall be a dry-type unit rated 1.6 MVA continuous, tested to ANSI C57.12.00.

Voltage Ratio and Impedance

The voltage ratio shall be 34.5 kV / 380 V, with 6.0% impedance on the self-cooled base.

Altitude and Temperature Derating

Ratings shall be guaranteed at 3,850 masl and 30 °C ambient, with the derating method stated.

Low Ambient Operation

The transformer shall operate without damage at -12 °C minimum ambient, with anti-condensation heaters.

Enclosure Humidity Protection

The enclosure shall prevent condensation up to 95% relative humidity and shall resist dust ingress.

Field Rectifier Arrangement

The field rectifier shall supply controlled direct current to the excitation windings of each drive motor.

Thyristor Bridge Configuration

A fully controlled thyristor bridge shall have redundant cooling fans and fused semiconductor arms.

Harmonic Performance

Rectifier harmonics shall be included in the drive assessment against IEEE 519-2014 at the point of common coupling.

Voltage Distortion Limits

Voltage distortion at the 34.5 kV bus shall not exceed 5.0% total or 3.0% individual.

Current Distortion Limit

Total demand distortion of current shall not exceed 5.0%, per the applicable IEEE 519-2014 band.

Interaction With Filter Bank

The vendor shall confirm that rectifier harmonics never overload filter bank 3110-FLT-01.

Protection and Monitoring

Protection against overcurrent, field loss, ground fault and winding overtemperature shall be provided, with alarms to 3210-LCP-01.

Primary Side Protection

The primary feeder from 3110-SWG-01 shall have relays coordinated with upstream devices and transformer inrush.

Temperature Monitoring Devices

Winding temperature detectors in each phase shall trip the supply on sustained overtemperature.

Seismic and Mechanical Design

The transformer, rectifier cubicles and anchorage shall suit Seismic Zone 4, 0.42 g peak ground acceleration and Site Class D.

Anchorage Calculations

Anchorage calculations shall be submitted showing that no component displaces or overturns under the design seismic event.

Testing and Quality Assurance

Routine tests shall follow ANSI C57.12.00, with manufacture under an ISO 9001:2015 certified quality system.

Impedance Verification Test

Measured impedance shall be compared with the guaranteed 6.0%, and deviations beyond standard tolerance shall be reported.

Certified Test Reports

Certified test reports for 3210-XFR-01 and the rectifier shall reach Ostrander Engineering Services Ltd. before shipment.

2.6 Cyclo-converter Transformers

This section covers cyclo-converter transformers 3210-XFR-02 and 3210-XFR-03, which supply the SAG and ball mill gearless drives. Each shall be designed, manufactured and tested to ANSI C57.12.00 for continuous service at 3,850 masl.

Transformer Ratings and Duty

Each cyclo-converter transformer shall be rated for continuous full-load drive operation under site conditions.

Drive Transformer Rated Capacity

Transformer 3210-XFR-02 shall be rated 20.0 MVA for 3210-MTR-01, and 3210-XFR-03 shall be rated 14.0 MVA for 3210-MTR-02.

Impedance and Tapping Requirements

Impedance and tapping shall limit drive fault duty and permit voltage adjustment at the drive input.

Short-Circuit Impedance Values

Both 3210-XFR-02 and 3210-XFR-03 shall have 7.5% impedance on the rated base, within ANSI C57.12.00 tolerances.

Impedance Matching Between Units

Both units shall have matched impedance to give consistent cyclo-converter source characteristics.

Tapping Range Provision

Transformer 3210-XFR-02 shall have taps of ±2 x 2.5%, and the vendor shall confirm the 3210-XFR-03 tapping.

Harmonic and Thermal Performance

The transformers shall carry cyclo-converter harmonic currents within permissible winding temperatures.

Harmonic Loading Design Basis

Winding and core design shall account for the drive harmonic spectrum to achieve IEEE 519-2014 compliance.

Distortion Limits at Coupling

Voltage distortion at the 34.5 kV point of common coupling shall not exceed 5.0% total or 3.0% individual.

Site Environmental Design Conditions

The transformers shall deliver rated output under Quebrada Azul site environmental conditions.

Ambient Temperature Range

Equipment shall suit ambient temperatures of -12 °C to 27 °C, with a 30 °C thermal design basis.

Altitude Effects on Insulation

Insulation and cooling shall be corrected for 3,850 masl, and the vendor shall state the correction.

Seismic Qualification and Anchorage

Transformers, bushings and anchorage shall withstand Zone 4 seismic loading of 0.42 g peak ground acceleration on Site Class D.

Testing and Quality Assurance

Each transformer shall be tested to ANSI C57.12.00 under an ISO 9001:2015 certified quality system.

Routine Factory Test Records

Routine test results shall be submitted to Ostrander Engineering Services Ltd. before shipment.

Documentation and Vendor Coordination

The supplier shall coordinate design data with Veltrane Drives GmbH to verify combined transformer and drive performance.

Nameplate and Tag Identification

Nameplates shall show tag, rating and impedance, with tags matching the project drawings.

Secondary Winding Data Submittal

The drive vendor shall define the secondary winding voltage and configuration in its design submittal.

Review Before Manufacture

The contractor shall accept the secondary winding data before any manufacturing release.

2.7 System Voltages

The gearless mill drive electrical system comprises a 23/34.5 kV medium-voltage bus, a 6.6 kV auxiliary bus and a 380 V low-voltage level. All equipment shall be rated for the applicable level.

Medium-Voltage Supply System

Both gearless drives shall be supplied at 34.5 kV as defined in document QAZ-5200-265-E-001_B.

Main Transformer Rating

3110-XFR-01 shall be rated 45 MVA ONAF and 30 MVA ONAN, 23 kV to 34.5 kV.

Main Transformer Impedance Requirement

The impedance of 3110-XFR-01 shall be 8.0%, confirmed by the vendor as compatible with switchgear 3110-SWG-01.

On-Load Tap Changer Range

3110-XFR-01 shall have an on-load tap changer providing ±10% in 17 positions, 1.25% per step.

Auxiliary Voltage System

The 6.6 kV auxiliary bus shall be supplied from 34.5 kV via 3110-XFR-02.

Auxiliary Transformer Characteristics

3110-XFR-02 shall be rated 6.0 MVA, 34.5 kV / 6.6 kV, 7.0% impedance, taps ±2 x 2.5%.

Auxiliary Motor Loads

Cooling-water pump motor 3220-MTR-01, 1,250 kW, shall connect to the 6.6 kV auxiliary bus.

Pump Motor Full-Load Current

Full-load current of 3220-MTR-01 at 6.6 kV shall be 129.4 A, at 0.88 power factor and 0.96 efficiency.

Motor Feeder Sizing Basis

Feeders and protection for 3220-MTR-01 shall be sized from this current, derated for 3,850 masl altitude.

Starting Voltage Performance

The 6.6 kV bus voltage shall remain acceptable during 3220-MTR-01 starting, as shown by the vendor's study.

Low-Voltage Distribution System

3220-XFR-01, rated 0.8 MVA, 6.6 kV / 380 V, 5.0% impedance, shall supply low-voltage loads.

Motor Control Center Supply

Motor control center 3220-MCC-01 shall serve 450 kW at 380 V, full-load current 855.7 A.

Load Current Calculation Basis

This current assumes 0.85 power factor and 0.94 efficiency; the incoming feeder shall be rated accordingly.

Drive Area Transformer Supply

Drive area transformers shall be fed at medium voltage, ratios confirmed by Veltrane Drives GmbH.

Excitation Transformer Rating

Excitation transformer 3210-XFR-01 shall be 1.6 MVA, 34.5 kV / 380 V, 6.0% impedance.

Cyclo-Converter Transformer Ratings

3210-XFR-02 and 3210-XFR-03 shall be rated 20.0 MVA and 14.0 MVA respectively, 7.5% impedance each.

Secondary Voltage Selection

The drive vendor shall select and state in the bid each cyclo-converter transformer secondary voltage.

SAG Drive Transformer Tap Range

3210-XFR-02 shall have taps of ±2 x 2.5% (5 positions) for supply voltage variation.

2.8 Mill Local Control Panel

The mill local control panel, tag 3210-LCP-01, shall provide local operation, indication and interlocking for SAG mill 3210-MLL-01 and ball mill 3210-MLL-02, and shall be supplied by Veltrane Drives GmbH.

Panel Power Supply Arrangement

The panel auxiliary supply shall be 380 V from 3220-MCC-01, fed through transformer 3220-XFR-01.

Supply Feeder Loading Basis

The panel supply shall be coordinated with the 450 kW total load of 3220-MCC-01.

Local Operator Controls

The panel shall provide local start, stop, inching and emergency stop for 3210-MTR-01 and 3210-MTR-02.

Control Location Selection

A key-operated selector shall transfer control between local and remote, with only one location commanding.

Emergency Stop Circuit Integrity

The emergency stop circuit shall be hardwired, fail-safe and independent of programmable logic in all modes.

Inching and Positioning Duty

Inching shall rotate each mill at reduced speed for liner maintenance under continuous push-button hold.

Indication and Display Requirements

The display shall indicate mill speed, drive current, drive status and active alarms in engineering units.

Speed Indication Ranges

Speed indication shall cover 10.13 rpm for the SAG mill and 11.29 rpm for the ball mill.

Current Indication Scaling

Current indication shall be scaled above full-load currents of 299.6 A for 3210-MTR-01 and 217.9 A for 3210-MTR-02.

2.9 Motor Control Center

The motor control center 3220-MCC-01 shall distribute low-voltage power to drive auxiliary loads in the 3220 auxiliary area. The assembly shall be rated for a connected load of 450 kW at 380 V.

Electrical Supply and Ratings

The motor control center shall be supplied at 380 V from auxiliary transformer 3220-XFR-01, rated 6.6 kV / 380 V.

Main Bus Current Rating

The main bus rating shall exceed the 855.7 A full-load current of 450 kW at 380 V.

Load Calculation Basis

Full-load current shall be calculated using a power factor of 0.85 and efficiency of 0.94.

Altitude and Temperature Derating

Ratings shall be derated for 3,850 masl and a 30 °C design ambient.

Control and Monitoring Interface

Status, trip and alarm signals from 3220-MCC-01 shall be provided to local control panel 3210-LCP-01.

Permissive and Interlock Signals

Cooling auxiliary starters shall provide run-confirmed contacts as gearless mill drive start permissives.

Environmental and Seismic Requirements

The assembly shall suit -12 °C to 27 °C ambient and 95% relative humidity.

Seismic Anchorage

Enclosures and anchorage shall withstand Zone 4 seismic loading, 0.42 g peak ground acceleration, on Site Class D soil.

Enclosure Mounting Details

Vertical sections shall be bolted together and floor-anchored for the specified seismic loads.

Anchor Calculation Submittal

The vendor shall submit seismic anchorage calculations before enclosure fabrication begins.

Quality and Code Compliance

Manufacture shall follow ISO 9001:2015, and installation shall comply with Norte Alto Electrical Code NAEC-2019.

2.10 Control System

This section defines control system requirements for gearless mill drives 3210-MTR-01 and 3210-MTR-02 at the Quebrada Azul Copper Expansion. The vendor shall provide integrated speed regulation, protection, sequencing, operator interfaces and plant communication.

Control Architecture Overview

The control system shall separate drive regulation, mill sequencing, and supervisory functions into hierarchical layers.

Drive Controller Hardware

Each drive shall have a dedicated controller within its enclosure, fed from an uninterruptible supply.

Controller Redundancy Provisions

Processor and power supply modules shall be redundant with automatic bumpless transfer.

Speed and Torque Regulation

The drive shall regulate speed and torque continuously within motor and mill limits.

Rated Operating Speed Targets

3210-MTR-01 shall run at 10.13 rpm and 3210-MTR-02 at 11.29 rpm, each 75.0% of critical speed.

Variable Speed Operation

Operators shall adjust mill speed within a vendor-defined range to suit ore and charge conditions.

Speed Setpoint Ramping

Speed setpoint changes shall follow adjustable ramps to limit mechanical stress on the mill shell.

Ramp Rate Limits

Ramp rate limits shall be configurable only at engineering level and recorded in the event log.

Torque Limitation Function

The torque limit shall protect against overload, with the SAG drive referenced to a rated torque of 15,549 kN·m.

Starting and Stopping Sequences

Start and stop sequences shall be automatic, each step requiring permissive feedback before proceeding.

Start Permissive Conditions

Starting shall be inhibited until lubrication, cooling water, excitation, and guard interlocks are healthy.

Cooling System Permissive

Permissives shall confirm cooling-water pump motor 3220-MTR-01 running with adequate flow.

Frozen Charge Protection

The system shall detect a frozen charge at start-up and stop the drive before it falls.

Frozen Charge Detection Method

Detection shall compare shell rotation angle with motor torque using a vendor-documented algorithm.

Charge Loosening Procedure

A controlled rocking procedure shall loosen a frozen charge within the motor torque limit.

Operator Authorisation Requirement

Charge loosening shall require operator authorisation from local control panel 3210-LCP-01 only.

Rocking Cycle Limits

The controller shall limit rocking cycles per attempt and apply a lockout at the limit.

Inching and Creep Operation

An inching mode shall rotate the mill slowly for liner replacement, inspection, and positioning.

Shell Positioning Accuracy

The drive shall position the shell to an entered angle and hold it by brake.

Protection and Interlocking

Protection shall operate independently of control logic so a control fault cannot disable a trip.

Electrical Protection Functions

Protection shall cover overcurrent, earth fault, over- and undervoltage, field loss, and stator overtemperature.

Overcurrent Reference Values

Overcurrent settings shall reference full-load currents of 299.6 A for 3210-MTR-01 and 217.9 A for 3210-MTR-02 at 34.5 kV.

Mechanical Protection Functions

Mechanical protection shall include air-gap, bearing temperature, and overspeed monitoring, each initiating a stop.

Emergency Stop System

A hardwired emergency stop shall remove torque and apply the brake regardless of controller state.

Emergency Stop Locations

Emergency stop pushbuttons shall be at the local panel, drive enclosure, and mill perimeter.

Reset After Emergency Stop

Emergency stop reset shall require manual action at the initiating station without automatic restart.

Operator Interface Requirements

Interfaces shall present status, alarms, and trends with consistent graphics at all stations.

Local Control Panel

Panel 3210-LCP-01 shall provide start, stop, inching, emergency stop, and a graphical display.

Control Location Selection

A selector shall assign control to the local panel or plant control system, never both.

Transfer Between Locations

Control transfer shall be bumpless and indicated at the local panel and central control room.

Plant Control System Interface

The drive control system shall exchange data, commands, and alarms with the plant control system.

Communication Network Design

Communication shall use redundant networks with fibre-optic links between buildings.

Signal Exchange List

The vendor shall prepare a signal exchange list giving each point's direction, data type, and units.

Hardwired Safety Signals

Safety signals, including emergency stop and trip status, shall also be hardwired.

Alarm Management and Prioritisation

Alarms shall be prioritised, time-stamped, and grouped by system to identify initiating causes.

First-Out Alarm Indication

The first-out alarm for each trip shall be displayed and retained until acknowledged and reset.

Event Recording and Logging

Alarms, trips, operator actions, and setpoint changes shall be logged with synchronised time stamps.

Clock and Time Synchronisation

Drive controller clocks shall synchronise with the plant master time source for event correlation.

Fault Diagnostics and Recording

Diagnostics shall capture high-resolution waveform data before and after every trip.

Transient Recorder Function

The transient recorder shall store currents, voltages, speed, and torque for local and workstation retrieval.

Power Quality Supervision

Supply quality at the point of common coupling shall be monitored and coordinated with harmonic mitigation.

Harmonic Distortion Monitoring

Voltage distortion shall be alarmed when exceeding the 5.0% total or 3.0% individual harmonic limit.

Harmonic Filter Coordination

Start-up shall require harmonic filter bank 3110-FLT-01 in service; its trip shall initiate a controlled stop.

Power Factor Control

Power factor shall not fall below 0.95 lagging at the 34.5 kV point of common coupling.

Reactive Power Coordination

Firing control shall be coordinated with filter switching to hold power factor during speed changes.

Transformer Tap Interaction

Control logic shall tolerate tap changer voltage steps of main transformer 3110-XFR-01 without nuisance tripping.

Tap Step Response

The controller shall ride through each 1.25% tap step with stable regulation.

Field Excitation Control

The controller shall regulate field excitation for optimum motor flux at all loads and speeds.

Excitation Supply Supervision

The drive shall trip on loss of supply from excitation transformer 3210-XFR-01 or field current.

Cyclo-Converter Firing Control

Firing control shall produce stator frequencies of 6.08 Hz for the SAG drive and 6.02 Hz for the ball drive.

Converter Transformer Monitoring

Each drive controller shall monitor signals from cyclo-converter transformers 3210-XFR-02 and 3210-XFR-03.

Transformer Alarm Levels

Winding and oil temperature alarm and trip levels shall follow the transformer manufacturer's recommendations.

Pre-Alarm Operator Notification

A pre-alarm shall precede any trip level so operators can reduce load.

Load Sharing Restrictions

The SAG and ball drives shall be controlled independently; a trip of one shall not trip the other.

Auxiliary Systems Control

Cooling, lubrication, and ventilation auxiliaries shall be controlled and supervised by the drive control system.

Auxiliary Power Supervision

The controller shall monitor motor control center 3220-MCC-01 and auxiliary transformer 3220-XFR-01.

Auxiliary Supply Loss Response

Auxiliary supply loss shall cause a controlled stop, with restart inhibited until restored.

Mill Lubrication Interlocks

Lubrication status shall be interlocked so the mill cannot rotate without adequate bearing lubrication.

High-Pressure Lift System

The high-pressure lift system shall be in service before starting and until running speed is reached.

Low-Pressure Lubrication Supervision

Loss of low-pressure lubrication flow shall alarm, then initiate a controlled stop after a vendor-defined delay.

Lubrication Temperature Limits

Oil supply temperature shall be controlled automatically for site temperatures down to -12 °C.

Brake System Control

The drive system shall control the brake and prevent motor torque against an engaged brake.

Brake Status Feedback

Brake positions shall be monitored by independent limit switches wired directly to the controller.

Environmental Design for Controls

Control equipment shall suit the 3,850 masl site altitude and project ambient conditions.

Ambient Temperature Range

Control panels shall operate from -12 °C to a design temperature of 30 °C.

Panel Climate Control

Panels shall include anti-condensation heaters and ventilation for relative humidity up to 95%.

Seismic Qualification of Panels

Panels and mountings shall withstand Zone 4 seismic loads with 0.42 g peak ground acceleration.

Control Software Requirements

Control software shall be structured, commented, and editable, without password-protected application code.

Software Version Control

The vendor shall version-control all software, identifying each release and its changes.

Software Backup Provisions

Full program and configuration backups shall be supplied at commissioning and after each modification.

Licence Transfer to Owner

All required software licences shall be transferred to Bramhall Copper Corp. at handover.

Access Control and Cybersecurity

Access shall be role-restricted and remote connections secured against unauthorised entry.

User Access Levels

Operator, maintenance, and engineering access levels shall be provided with individual user credentials.

Remote Diagnostic Access

Remote diagnostic access by Veltrane Drives GmbH shall require local enabling by site personnel.

Remote Session Logging

Remote sessions shall be logged with user, start time, and duration, and ended automatically when idle.

Control System Documentation Requirements

The vendor shall supply documentation sufficient for operation, maintenance, and modification of the drive controls.

Functional Description Document

A functional description of all modes, sequences, interlocks, and alarms shall be reviewed before coding.

Factory Acceptance Testing

The control system shall be factory tested with simulated field signals before shipment to Quebrada Azul.

Test Procedure Submission

Test procedures shall be approved by Ostrander Engineering Services Ltd. before the factory test.

Witness Test Notification

The vendor shall give advance written notice of factory tests for witness attendance.

Plant Interface Testing

Plant communication shall be factory demonstrated using the approved signal exchange list.

Site Acceptance and Commissioning

Site commissioning shall verify each control function from energisation through loaded mill operation.

Protection Function Verification

Every protective trip and interlock shall be site tested and recorded on signed commissioning sheets.

Harmonic Compliance Testing

Harmonic measurements during loaded commissioning shall confirm compliance with this specification's limits.

Current Distortion Verification

Measured current total demand distortion shall not exceed 5.0%, assessed against IEEE 519-2014.

Operator and Maintenance Training

The vendor shall train operators and maintenance personnel in operation, fault response, and diagnostics.

Training Materials and Manuals

Training materials shall be drive-specific and delivered in printed and electronic form.

Spare Parts and Support

The vendor shall recommend spare parts and guarantee replacement availability for a stated support period.

Commissioning Spare Parts Provision

Commissioning spare modules shall be held at site by the vendor, separate from operating spares.

Remote Site Logistics

Spare holdings shall suit the site near Calderón Verde, 620 km from the regional capital.

Hardware Obsolescence Management

The vendor shall notify planned hardware obsolescence in writing and propose migration paths.

Quality Assurance Requirements

Design, manufacture, and testing shall follow a quality system certified to ISO 9001:2015.

Local Electrical Code Compliance

Panel construction and wiring shall comply with the Norte Alto Electrical Code NAEC-2019.

Inspection Records Retention

Inspection and test records shall be retained and available to Bramhall Copper Corp. on request.

2.11 Electrical Room Package

This section defines requirements for the prefabricated electrical room housing the drive equipment supplied by Veltrane Drives GmbH. The package shall be suitable for continuous operation at 3,850 masl in Norte Alto Province.

Scope of Room Package Supply

Supply shall include the room structure, electrical services and drive control enclosures, including panel 3210-LCP-01.

Equipment Housed Within Room

The room shall house the cyclo-converter and excitation equipment for motors 3210-MTR-01 and 3210-MTR-02, with maintenance clearance.

Interfaces With Other Packages

Cable entries, earthing and control interfaces shall be coordinated with 34.5 kV switchgear 3110-SWG-01 and the drive transformers.

Excluded Items and Battery Limits

Vendor drawings shall show battery limits, and the bid shall clearly identify excluded items.

Structural and Seismic Design

The structure shall be designed for seismic Zone 4, 0.42 g peak ground acceleration and site class D.

Wind Loading on Room Structure

The structure shall resist a 36 m/s design wind speed, as a 3-second gust at 10 m.

Environmental Design Conditions

The package shall be rated for ambient temperatures from -12 °C to 30 °C.

Heating, Ventilation and Air Conditioning

HVAC shall maintain drive equipment limits at outdoor relative humidity of 95%.

Altitude Correction of Cooling Equipment

Cooling capacity shall be corrected for reduced air density at 3,850 masl.

Redundancy of Cooling Units

Failure of any single cooling unit shall not cause a drive trip or derating.

Dust Exclusion and Pressurization

The room shall be kept at positive pressure, with all intake air filtered for site dust.

Internal Power Distribution

Room services shall be supplied at 380 V from 3220-MCC-01 via transformer 3220-XFR-01.

Lighting and Small Power

Normal and emergency lighting and convenience outlets shall comply with Norte Alto Electrical Code NAEC-2019.

Earthing and Bonding Provisions

An internal earth bus bonded to the site grid shall connect every enclosure and cable tray.

Fire Detection and Protection

Fire detection and suppression shall be acceptable to the Local Authority Having Jurisdiction.

Factory Assembly and Testing

The package shall be fully assembled and tested at the vendor works under ISO 9001:2015.

Transport and Site Installation

The package shall be split into sections for road transport over the 620 km route.

Documentation for Installation and Commissioning

Installation, lifting and commissioning documents shall be submitted before shipment and reference QAZ-EP-0000-265-SPC-00006-Q.

2.12 Harmonics and Power Factor Study

The drive vendor shall perform a harmonic and power factor study demonstrating IEEE 519-2014 compliance at the 34.5 kV point of common coupling and confirming harmonic filter bank adequacy under all operating conditions.

Study Scope and Responsibility

Veltrane Drives GmbH shall study both mill drives and filter bank 3110-FLT-01.

Network Model Boundaries

The network model shall extend from the 23 kV supply through 3110-XFR-01 to every downstream bus.

Harmonic Distortion Limits

Distortion at the point of common coupling shall meet the following limits.

Voltage Distortion Criteria

Voltage THD shall not exceed 5.0%, nor any individual harmonic 3.0%, at the 34.5 kV bus.

Current Distortion Criteria

Current total demand distortion shall not exceed 5.0%, per IEEE 519-2014 for a short-circuit ratio below 20.

Interharmonic and Sideband Evaluation

The study shall evaluate cyclo-converter interharmonics over the full speed range, including inching.

Power Factor Requirements

Power factor shall be at least 0.95 lagging at the 34.5 kV point of common coupling during normal grinding.

Drive Load Basis

SAG drive 3210-MTR-01 (16,500 kW) and ball drive 3210-MTR-02 (12,000 kW) shall form the reactive load basis.

Input Apparent Power Values

Input demands of 17,906 kVA and 13,022 kVA shall be used, based on 0.95 power factor and 0.97 efficiency.

Light Load Operating Cases

Single-mill and reduced-load cases shall confirm the power factor criterion is maintained.

Harmonic Filter Bank Design

Filter bank 3110-FLT-01 shall be rated 6.0 Mvar at 34.5 kV, tuned to the 5th harmonic.

Detuning and Resonance Checks

The study shall identify parallel resonances and confirm capacitor tolerance drift causes no detuning.

System Data and Assumptions

The study shall adopt the transformer impedances, tap ranges and site conditions specified herein.

Main Transformer Parameters

3110-XFR-01 shall be modelled with 8.0% impedance and a ±10%, 17-position on-load tap changer.

Converter Transformer Parameters

Transformers 3210-XFR-02 and 3210-XFR-03 shall be modelled at 20.0 MVA and 14.0 MVA, each with 7.5% impedance.

Altitude and Ambient Effects

Study ratings shall reflect the 3,850 masl site altitude and the 0.94 motor derating factor.

Study Deliverables and Review

The study report shall be submitted for review before filter bank design release.

Report Content

The report shall present the network model, harmonic spectra, impedance scans, power factor results and a per-case compliance summary.

Field Verification Measurements

Commissioning measurements shall verify the harmonic and power factor predictions.

2.13 Special Tools and Spare Parts

This section 2.13 Special Tools and Spare Parts defines Veltrane Drives GmbH obligations for special tools and spares for gearless drives 3210-MTR-01 and 3210-MTR-02. Tool functionality shall be demonstrated during 3.1 Factory Inspection and Tests.

Special Tools and Lifting Devices

The Vendor shall supply all special tools to install, align, maintain and dismantle both mill drives.

Air Gap Measurement Tools

Each motor shall be provided with air gap measurement and adjustment instruments with traceable calibration certificates.

Pole Removal and Handling Fixtures

Fixtures shall permit rotor pole removal on 3210-MTR-01 and 3210-MTR-02 without dismantling the stator frame.

Fixture Load Rating Documentation

Each lifting fixture shall carry a load rating plate, with design calculations submitted before fabrication.

Commissioning Spare Parts

The Vendor shall furnish commissioning spares sufficient to complete energisation and performance testing of both drives.

Unused Commissioning Spares Return

Unused commissioning spares shall become property of Bramhall Copper Corp. and be transferred to site stores.

Operating and Capital Spares

The Vendor shall propose spares lists for the drives, transformers 3210-XFR-01, 3210-XFR-02, 3210-XFR-03 and panel 3210-LCP-01.

Spares Suitability for Site Altitude

All spares shall be rated for 3,850 masl and ambient temperatures from -12 °C to 30 °C.

Electronic Spares Storage Conditions

Electronic spares shall be packed in sealed, humidity-controlled containers suitable for relative humidity up to 95%.

Shelf Life and Revision Control

Each electronic spare shall show its hardware revision and shelf-life limit, matching the in-service revision.

Spares Lists Submittal and Interchangeability

Spares lists shall identify part numbers, interchangeability between 3210-MTR-01 and 3210-MTR-02, and stocking, and shall be submitted with the bid.

3 EXECUTION

This section defines execution requirements for the gearless mill drives, following the supply provisions ending with 2.13 Special Tools and Spare Parts. It shall govern inspection, testing, documentation and acceptance by Veltrane Drives GmbH before shipment.

3.1 Factory Inspection and Tests

Clause 3.1 Factory Inspection and Tests establishes the tests Veltrane Drives GmbH shall perform for 3210-MTR-01 and 3210-MTR-02. Bramhall Copper Corp. and Ostrander Engineering Services Ltd. may witness, and nothing shall ship until results are accepted.

General Test Requirements

Factory tests shall be performed under an ISO 9001:2015 quality system using traceably calibrated instruments.

Inspection and Test Plan

An inspection and test plan defining hold and witness points shall be approved before manufacture.

Witness Notification Requirements

The vendor shall give timely written notice of each witness point to permit Purchaser attendance.

Gearless Motor Stator Tests

Stator segments for 3210-MTR-01 and 3210-MTR-02 shall be tested to IEC 60034-1 before dispatch.

Winding Insulation Verification

Each stator segment winding shall receive recorded insulation resistance, polarization index and withstand tests.

Stator Core Integrity Check

Each segment shall receive a core loss or flux test to detect interlaminar faults before winding.

Altitude Correction of Test Voltages

Dielectric test levels shall account for 3,850 masl operation and the 0.94 motor derating factor.

Rotor and Pole Tests

The 72 SAG drive poles and 64 ball drive poles shall each be individually inspected and tested.

Pole Winding Resistance Measurement

Each pole winding's direct current resistance shall be measured and corrected to a reference temperature.

Cyclo-Converter Factory Tests

Each cyclo-converter shall undergo routine functional tests of firing control, protection and cooling before shipment.

Converter Light Load Testing

Converter sections shall be energized at reduced load to confirm thyristor operation, gate pulses and current sharing.

Harmonic Performance Verification

Test records shall demonstrate compliance with IEEE 519-2014 limits of 5.0% voltage THD and 5.0% current TDD.

Control System Integration Test

The drive control system, including 3210-LCP-01, shall be integration tested with simulated mill signals.

Drive Transformer Tests

Transformers 3210-XFR-01, 3210-XFR-02 and 3210-XFR-03 shall be routine tested to ANSI C57.12.00.

Impedance and Loss Measurement

Measured impedance shall be compared against 6.0% for the excitation transformer and 7.5% per cyclo-converter transformer.

Tap Changer Operation Check

The 3210-XFR-02 tap changer ratio shall be measured at each of its 5 positions within the ±2 x 2.5% range.

Ratio Test Acceptance Records

Ratio deviations at every tap shall be reported against the ANSI C57.12.00 tolerance.

Witnessed Acceptance Testing

Bramhall Copper Corp. may witness any factory test, with vendor access provided to all test areas.

Purchaser Representative Responsibilities

Ostrander Engineering Services Ltd., as Purchaser's representative, shall sign satisfactorily witnessed test records.

Retest After Failure

Failed items shall be repaired or replaced and fully retested under the original witness arrangements.

Root Cause Investigation Report

The vendor shall report each failure's root cause, corrective action and effect on identical units.

Test Documentation and Reports

Certified test reports shall reference QAZ-EP-0000-265-SPC-00006-Q and the applicable equipment tag.

Report Content Requirements

Reports shall state test and ambient conditions, instruments, measured values, acceptance criteria and witness signatures.

Preparation for Shipment

Accepted equipment shall be cleaned, preserved and packed for transport to Calderón Verde and site.

Preservation for Site Conditions

Packing shall protect against site temperatures of -12 °C to 27 °C and 95% relative humidity.

Long Term Storage Provisions

For site storage, the vendor shall provide written instructions covering space heaters and periodic insulation checks.

3.2 Site Commissioning and Operator Training

This Section 3.2 Site Commissioning and Operator Training defines the commissioning obligations of Veltrane Drives GmbH for the Quebrada Azul gearless mill drives and the training of Bramhall Copper Corp. personnel before handover.

Commissioning Organization and Responsibilities

The vendor shall provide a qualified site commissioning team coordinated with Ostrander Engineering Services Ltd.

Commissioning Lead Engineer

The vendor shall nominate a commissioning lead engineer experienced on gearless drives of comparable rating.

Site Safety and Isolation

Energization of 3110-SWG-01 and downstream equipment shall follow Local Authority Having Jurisdiction isolation procedures.

Permit to Work Procedures

No circuit shall be energized without a signed permit, with lockout per Norte Alto Electrical Code NAEC-2019.

Pre-Energization Installation Checks

Before energization, the vendor shall verify 3210-MTR-01 and 3210-MTR-02 installation, wiring, cooling and protection settings.

Transformer Inspection and Testing

Transformers 3210-XFR-01, 3210-XFR-02 and 3210-XFR-03 shall be site tested per ANSI C57.12.00 before energization.

Energization and Functional Testing

Drives shall be energized sequentially, starting with auxiliaries fed from 3110-XFR-02 and 3220-XFR-01.

Auxiliary System Verification

Cooling-water pump motor 3220-MTR-01 and 3220-MCC-01 shall be proven before any mill drive rotation.

Uncoupled and Inching Runs

Each motor shall first be inched from 3210-LCP-01 to confirm rotation direction and protection response.

Rated Speed Verification

The vendor shall demonstrate 10.13 rpm for 3210-MLL-01 and 11.29 rpm for 3210-MLL-02, 75.0% of critical speed.

Load Performance Testing

Under process load, power factor shall be 0.95 lagging or better at the 34.5 kV point of common coupling.

Power Quality Compliance

Harmonics shall be measured at the point of common coupling with 3110-FLT-01 and both drives operating.

Harmonic Distortion Limits

Voltage THD shall not exceed 5.0%, individual voltage harmonics 3.0%, and current TDD 5.0%.

Harmonic Measurement Reporting

The vendor shall submit a harmonic test report showing IEEE 519-2014 compliance before acceptance for continuous operation.

Operator Training Program

The vendor shall deliver structured operations and maintenance training for Bramhall Copper Corp. personnel.

Classroom Instruction Content

Classroom sessions shall cover drive principles, cyclo-converter operation, protection, alarm handling and safe isolation.

Hands-On Field Training

Field training on installed equipment shall include start, stop and fault recovery at 3210-LCP-01.

Training Materials and Language

Training manuals shall be in the project language and reflect the as-commissioned drive configuration.

Completion and Handover

Commissioning shall be complete when all tests are accepted and safety-affecting punch list items are closed.

Commissioning Records Submission

The vendor shall submit signed test records, final protection settings and training attendance per ISO 9001:2015.

3.3 Product Delivery and Handling

Section 3.3 Product Delivery and Handling defines packing, transport, receipt, storage and handling requirements for gearless mill drive equipment supplied by Veltrane Drives GmbH. All equipment shall arrive at site fit for installation.

Packing and Preservation Requirements

All drive components shall be packed to withstand transport without damage, moisture ingress or corrosion.

Motor Stator and Rotor Packing

Stator segments and rotor poles of 3210-MTR-01 and 3210-MTR-02 shall be crated, braced, sealed and desiccated.

Shock and Tilt Indicators

Each motor crate shall carry shock and tilt indicators, and triggered indicators shall be recorded at receipt.

Transformer Shipping Provisions

Transformers 3210-XFR-02 and 3210-XFR-03 shall ship with impact recorders and a Vendor-specified gas or oil fill.

Transport to Site

Routes to Calderón Verde and the site shall be surveyed for clearances, axle loads and gradients.

High Altitude Route Considerations

The Vendor shall account for the 3,850 masl site altitude and the 620 km distance from the regional capital.

Cold Weather Transit Protection

Equipment in transit shall be protected against temperatures down to -12 °C and against condensation.

Heater Supply During Transit

The Vendor shall state whether fitted space heaters need energizing during transit or storage.

Receipt and Inspection

Ostrander Engineering Services Ltd. shall inspect deliveries against packing lists and record damage before acceptance.

Documentation Accompanying Each Shipment

Each shipment shall carry packing lists, preservation instructions and tag references consistent with QAZ-EP-0000-265-DST-00006-Q.

Damage and Discrepancy Reporting

Damage or shortage shall be reported in writing to Bramhall Copper Corp., and the Vendor shall propose corrective action.

Witnessed Unpacking of Motors

The Vendor representative shall witness unpacking of motor stator segments to confirm condition and preserve warranty.

Site Storage Requirements

Stored equipment shall be protected from site conditions, including relative humidity up to 95%, in coordination with QAZ-EP-0000-210-DBD-00002-B.

Indoor Storage of Electronics

Cyclo-converter, control and excitation equipment, including 3210-LCP-01, shall be stored indoors in dry, heated, dust-free conditions.

Lifting and Handling

Lifting shall follow the Vendor handling drawings, using only designated lifting points and spreader arrangements.

Heavy Lift Planning

Heavy lift plans shall be submitted for review and shall reference QAZ-EP-3100-255-DST-55017-Q where applicable.

Wind Limits for Lifting

Lifting shall stop as wind approaches lift plan limits, recognizing the 36 m/s site design wind.

Seismic Restraint During Storage

Stored heavy packages shall be restrained against overturning for seismic Zone 4 and 0.42 g peak ground acceleration.

Referenced Handling Documents

Handling and storage shall be consistent with QAZ-5200-265-E-001_B and the items in Section 4 ATTACHMENTS.

Handover to Installation

Preservation records shall be handed to the installation team so preservation continues until energization.

4 ATTACHMENTS

The following documents are attached to this specification and shall be read with it: QAZ-EP-0000-210-DBD-00002-B, Site Conditions; QAZ-EP-0000-265-DST-00006-Q, Gearless Mill Drive Datasheet; QAZ-EP-3100-255-DST-55017-Q, Grinding Mill Data Sheet; and QAZ-5200-265-E-001_B, Single Line Diagram. Where an attachment conflicts with this specification, the drive vendor shall refer the matter to the contractor for resolution before proceeding.